Masoud Hemmatpour

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Nationality: Iranian

Date of Birth: May 29, 1983

Residence:Tromso, Norway

Gender: Male

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Mission statement

Dedicated professional with a profound passion for shaping the future of technology through the integration of in-network computing, machine learning, and intelligent edge solutions. Committed to driving innovation, I leverage my expertise to architect and implement cutting-edge systems that seamlessly blend emerging technologies.

Positions

- **Arctic University of Norway (UiT)** — Tromso, Norway
  Postdoctoral Fellow — *May. 2023 – Current*
  - **Project**
    - In-network computing in IoT systems
- **Simula Lab** — Oslo, Norway
  Adjunct Research Scientist — *May. 2023 – Dec. 2023*
  - **Project**
    - In-network computing in HPC systems
- **Simula Lab** — Oslo, Norway
  Postdoctoral Fellow — *Mar. 2021 – May. 2023*
  - **Project**
    - In-network computing in HPC systems
- **Cisco systems** — Paris, France
  Senior Software engineer — *Jan. 2019 – Jan. 2020*
  - **Project**
    - Control and management plane in cloud-native environment

Education

- **Politecnico di Torino** — Torino, Italy
  PhD in computer engineering — *Nov. 2015 – Sept. 2019*
  - **Thesis**
    - High Performance Computing using InfiniBand based clusters
- **Politecnico di Torino** — Torino, Italy
  MSc in communications and computer networks engineering — *Oct. 2012 – Oct. 2015*
  - **Thesis**
    - Read-copy update synchronization technique improvement by hardware message passing
      Research project at École Politechnique Fédérale de Lausanne (EPFL)
- **Azad university of najaf abad** — Najaf abad, Iran
  Bachelor of Science, Computer Software Engineering — *2005 – 2008*
  - **Thesis**
    - Implementing bulk SMS (Short Message Service) sender with AT commands in Delphi programming
- **Bu-Ali Sina University** — Hamedan, Iran
  Association Degree, Computer Software — *2002 – 2004*
  - **Thesis**
    - Implementing answering machine with TAPI , Delphi programming

Work experience

- In-network Machine Learning (European Union's Horizon MISO project - University of Tromso)
  - The goal of MISO project is to develop and demonstrate an autonomous in-situ observation platform for use in hard-to-reach areas. I develop an intelligent framework to reduce the power consumption of end nodes. Moreover, I develop an in-network intelligent solution to detect anomalies of IoT nodes.
- In-network computing (Simula Research Laboratory)
  - Exploring in-network computing research directions.
  - working on several programmable network devices including FPGA and Soc based SmartNIC as well as programmable switch.
- Network management (Linux Foundation Collaborative Project in Cisco Systems)
  - Designing and implementing a Yet Another Next Generation (YANG) model to enable VPP to communicate with remote controllers such as OpenDaylight (ODL) and Network Services Orchestrator (NSO) in order to receive telemetry and push configuration.
  - Enabling VPP to communicate with a network routing software suite such as FRRouting (FRR).
- Remote Direct Memory Access (RDMA) (Research project Politecnico di Torino & T.J. Watson IBM research center)
  - Investigating performance challenges of RDMA operations in InfiniBand based clusters.
  - Designing and implementing an RDMA enabled in-memory key-value store.
- Health care system (OPportunities for active and healthy LONgevity (OPLON) project)
  - Designing and implementing a holistic approach to detect an abnormal gait in order to avoid an unintentional fall and in case of a fall detection reduces injuries and notifies care givers.
- Fault injection (Research Project Politecnico di Torino & University of Montpellier)
  - Designing and implementing a framework to automatically inject faults into a real-time operating system (i.e., FreeRTOS) for STM32 Discovery boards.
- Linux kernel scheduler optimization (Research Project at Politecnico di Torino)
  - Introducing new parameter (i.e., the number of data cache misses) at the given time slice to the Completely Fair Scheduler (CFS). In this way, the scheduler can penalizes processes which waste CPU cycles to handle cache miss instead of the main task.
- Synchronization algorithm on multi core architecture (Research Project at Politecnico di Torino)
  - Implementing a framework to evaluate different synchronization methods including non-blocking algorithms.
- Urban air quality (Cyclair project Torino Living Lab)
  - Investigating the relevance of height in dust monitoring systems based on Libelium Waspmote connected to a dust sensor OPC-N2.
- System-on-a-chip (Research project Politecnico di Torino & École polytechnique fédérale de Lausanne (EPFL))
  - Exploiting hardware message passing feature in system-on-a-chip CPU (i.e., TILE-Gx36) in order to enhance the performnce of the non-blocking synchronization algorithm.
- Co-founder Fanavaran Pasargad J, Linux Professional Institute (LPI) partner
  - Teaching LPI courses to universities and organizations

Submitted Proposal

- NetCrush: Crushing Communication wall with In-Network Machine Learning (Excellence: 5/7)

- Smart RDMA: In-Network Accelerated Processing of Big Data (Excellence: 6/7)

Teaching Assistant

- Teaching assistant in System Programming course, Politecnico di Torino, 3 academic years
- Teaching assistant in Operating Systems course, Politecnico di Torino, 2 academic years
- Teaching Linux Professional Institute courses in several universities and companies

Visiting programs and talks

- Laboratory for Information, Networking and Communication Sciences (LINCS) 2023
- University of Oxford 2023
- University of California Davis (UC DAVIS) 2018
- Purdue University 2017
- École Polytechnique Fédérale de Lausanne (EPFL) 2015

Awards

- Best Dissertation Award Politecnico di Torino 2019
- Quality Award Politecnico di Torino 2018
- Quality Award Politecnico di Torino 2017

References

Mohammad Sadoghi
Associate professor at the department of computer science University of California Davis, USA
Email: mo.sadoghi@expolab.org

Amir Taherkordi
Associate professor at the Department of Informatics University of Oslo, Norway
Email:amirhost@ifi.uio.no